Autism Spectrum Disorders

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Introduction

Autism spectrum disorder (ASD) is more commonly diagnosed in recent times. "Spectrum" in ASD indicates that each individual is affected in different ways, with mild-to-severe symptoms often with overlapping comorbidities.

Definition

Autism spectrum disorder is a biologically based neurodevelopmental disorder with core deficits in two domains: Social communication/interaction and restrictive, repetitive patterns of behavior.

Clinical Features

Symptoms appear before 3 years of age but may be obvious over time when social demands exceed limited capacities **(Box 1)**.

BOX 1: Red flags for autism.

- ☑ Poor response to name call by 1 year of age
- ☑ No meaningful words by 18 months of age
- ☑ Does not play with toys appropriately
- ☑ Fixed patterns of interest/activities
- ☑ Decreased interaction with peers/friends
- ☑ Has odd movement patterns and/or very repetitive behaviors
- ☑ Demonstrates echolalia (repeats words)
- ☑ Is hyperactive
- ☑ Toe walking

Coexisting Conditions

Conditions co-occurring with ASD are intellectual disability, attention deficit/hyperactivity disorder (ADHD), sleep disorders, feeding and gastrointestinal symptoms, seizures, obesity, anxiety, and wandering.

Differential Diagnosis

- ☑ **Hearing loss:** The child has normal social interaction and nonverbal communication, lacking only verbal communication.
- ☑ **Selective mutism:** The child is completely mute only in the presence of strangers.
- ☑ **Global developmental delay/intellectual disability:** Development is globally affected. Social communication is affected on par with the other domains.
- ☑ **ADHD:** Social communication domain is not significantly affected. Hyperactivity is usually meaningful. Intelligence is not affected.

Screening

Children between 16 and 30 months should be screened with the M-CHAT-R/F combined with clinical evaluation.

Diagnosis

No single test/tool is appropriate for all children; hence, a comprehensive evaluation by a team led by a developmental behavioral pediatrician is necessary for proper diagnosis. This includes a combination of:

- ☑ **History and physical examination**
- ☑ Tools such as:
  - **Measures relying on parent report:** The Autism Behavior Checklist (ABC), Gilliam Autism Rating Scale, 3rd edition (GARS-3), and the Autism Diagnostic Interview-Revised (ADI-R)
  - **Measures using direct observation:** The Childhood Autism Rating Scale (CARS) and the Autism Diagnostic Observation Schedule-Generic (ADOS-2)
- ☑ Developmental/intelligence testing with separate estimates for verbal and nonverbal skills; assessment of adaptive skills to document the presence of associated intellectual disability.
- ☑ Sensorimotor and/or occupational therapy evaluation, speech, language, and communication assessment

Investigations

- ☑ Hearing, vision, complete blood count (CBC), and thyroid function test are recommended.
- ☑ Routine metabolic testing, biomarkers, and environmental toxins are not recommended for regular use; lead levels are indicated only with definite history of exposure.
- ☑ **Neuroimaging: Indications:** ASD with:
  - Abnormal neurologic examination
  - Seizures
  - Headache
- ☑ **Electroencephalogram**—*sleep-deprived electroencephalogram (EEG):* Indicated when:
  - History of unusual spells or behaviors frankly suggestive of seizures
  - *History of significant regression in language skills*: Landau–Kleffner syndrome (acquired epileptic aphasia)

Investigations

- ☑ **Genetic evaluation:** Recommended especially in the presence of dysmorphic features or intellectual disability.
  - No specific mutation has been identified that is specific for ASD.
  - Recommended to test for fragile X syndrome for all children with ASD, especially for boys and children with a suggestive family history of male members with intellectual disability.
  - Chromosomal microarray (CMA) is recommended, if the etiology is not known; it reveals a pathogenic copy number variant (CNV) in approximately 9% cases.
- ☑ If history and clinical examination, CMA, and fragile X analysis do reveal the etiology, it is recommended to advise whole exome sequencing (WES), which allows for the identification of single-nucleotide variants.
- ☑ **Genetic counseling:** The risk of ASD in the next pregnancy is 10% greater with one child and about 34% with two or more previous children with idiopathic ASD.

Management

- ☑ At least a provisional diagnosis is essential to enable appropriate intervention. Assessment of comorbidities and ruling out differential diagnosis is key to a good outcome.
- ☑ No single therapy has proven to be especially effective; robust evidence for treatment programs in children younger than 2 years and in adolescents is still emerging.
- ☑ Intervention should target core features of autism, i.e., deficits in social communication and interaction, and restricted repetitive patterns of behavior, activities, and/or interests rather than symptoms such as "delayed speech" and "hyperactivity."
- ☑ Intervention should be evidence-based, structured, and appropriate to the developmental and functional needs of the child in association with the parents and family. Intervention should have a larger component of home and community-based interventions rather than merely clinic-based.
- ☑ Management of co-morbidities such as intellectual disability, ADHD, anxiety, and depression is critical to effectiveness of treatment.
- ☑ Pharmacotherapy may be offered to children when there is a specific target symptom or comorbid condition. Methylphenidate and atomoxetine are used to decrease hyperactivity and increase attention span, risperidone for aggressive behavior, and selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine or sertraline for anxiety and repetitive behaviors. Melatonin may be used for sleep disorders. Pharmacologic management of seizures and gastrointestinal disorders in children with ASD are like that in children without ASD.
- ☑ Treatment programs should be monitored to ensure appropriate response to therapy. The program should be reviewed and modified, as the child's abilities and needs change over time.

Management

- ☑ The role of the pediatrician is in early suspicion, counseling of parents, appropriate referral, and most important, regular monitoring to note improvements in the child from time to time. Blanket statements such as "your child can never be normal" should be avoided—parents end up either refusing intervention or with decades of inefficient and repetitive or complementary interventions.
- ☑ Nongoal-oriented, arbitrary, unending, repetitive, symptom-based interventions should be monitored for their effectivity versus their potential to harm the child as well as take up the golden early period of life which has maximal potential for improvement, and cautioned against if needed.
- ☑ With appropriate intervention, most children with mild-to-moderate ASD, especially in the absence of severe comorbidities, should be steered toward normalcy and meaningful social inclusion.

Myths and Facts

- ☑ *Vaccination*: There is clear evidence that vaccines do not cause ASD.
- ☑ *Dietary changes and use of other modalities*: Gluten-free, casein-free diet is not recommended for children with ASD unless they have celiac disease or gluten sensitivity.
- ☑ Similarly, complementary and alternative therapies including antimicrobial agents, intravenous immunoglobulin, vitamin A, vitamin B_6 and magnesium, chelation, hyperbaric oxygen, homeopathy, vagus nerve stimulation, and stem cell transplantation are not recommended.

Conclusion

Though early intervention is the buzz word, it is equally important to have the correct, appropriate, and dynamic intervention, based on a documented holistic program/plan responsive to improvements/worsening in the child, rather than arbitrary or blanket referrals to rigid symptom-based approaches.

Further Reading

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